

Modified notes on a set theory course



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Introduction

These notes, that I won't consider a book because of its informality, were born as an idea after ending the course, Álgebra IV in ESFM IPN, unsatisfied. That isn't saying that I hated Dr. Hugo's classes. I love him. But the contents were not the best for my idea of an algebra IV course focused on set theory.

These notes are meant to help understand and clarify a lot of things not included in the course I took. Is not some kind of tortuous book that some gifted individuals can read, I'm not that good. It is written in a way that makes the part of the course almost completely independent of the first chapter. However, for a better understanding on what set theory is constructed over, it should be read at least once or twice because my writing, as a lot of people can confirm, is not the best.

The plan is, to clarify things about logic, especially for first order logic that is what set theory is usually developed over. But also, it works as prerequisites for further topics on set theory and what interests me most, foundations of mathematics. Next chapter deals with the usual course on set theory, giving axioms and its interpretations, proving statements about relations, functions, equivalence, unicity, existence, etc. The final stop for this chapter are the real numbers and its properties constructed via sets. A further study on set theory, topics like cardinality, axiomatical systems and combinatorics are, aside from interesting, sometimes very necessary for further topics on algebra, analysis, etc. Not indispensable, yes, but really good to know for them. A chapter about structures is then introduced, trying to get to universal algebra and an introduction to category theory via set theory. And finally, a chapter on the foundations for the curious mind that wants to waste some time on interesting things like foundational category theory, drug addict thinking in metamathematics and a formalization of logic via mathematics, a seemingly circular foundation. Also, with sections that are of my interest, like type theory and philosophy of mathematics.

All of this, in my opinion, conforms an overloaded course on set theory, and should be taken very selectively and with a grain of salt. I'm just a but a dumb student right now.



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Logic

1.1 First order formal languages

Every definition given here might be understood as not strictly mathematical. That meaning, we won't dive too much into detailed mathematical descriptions nor ontological remarks about the things that we define. Collections are to be understood in an intuitive sense, as well as known collections like the natural numbers. Consider the collection of natural numbers given along with its operations $+$, $\cdot$.

Definition. A *first order formal language* $\mathcal{L}$ is a collection of symbols, each identified with a type¹. It can be interpreted as a collection of collections.

1. At most an infinite countable² amount of variables:

$$x_1, x_2, x_3, \dots$$

2. Constant symbols:

$$c_1, c_2, c_3, \dots$$

3. Relation symbols, each of them with an associated range n_i , a natural number greater than 0. That should be interpreted as “they relate n_i variables/constants”:

$$R_1^{n_1}, R_2^{n_2}, R_3^{n_3}, \dots$$

4. Function symbols, each of them also with a range m_i :

$$f_1^{m_1}, f_2^{m_2}, f_3^{m_3}, \dots$$

5. Logical connectives:

$$\rightarrow, \neg, \vee, \wedge, \leftrightarrow.$$

6. Quantifiers:

$$\forall, \exists.$$

7. Parenthesis and comma:

$$(,), , .$$

Function symbols with range 0 are to be thought of as constant symbols. In intuition, they don't take any input and just output the same thing always. Syntactically, they never change its written form. We also call a range n function symbol a n -ary function symbol, the same holds for relations. We shall use the usual language after doing this remark, i.e. unary for 1-ary, binary for 2-ary, etc.

What the “first order” part of the definition actually means will be determined later, but for an intuitive idea, it refers to the “level” on which the things we talk about reside. For example, to make a statement about natural numbers, we should use the first order language of arithmetic. To declare induction on natural numbers we would need to talk about statements of natural numbers, that is, to use a higher order language, in this case, second order³.

¹As in, this object is of type “Natural number”, or of type “Logical connective”.

²Each variable can be related one-to-one to a natural number.

³But is not strictly necessary, as we may see further into these notes.

Definition. For a first order formal language $\mathcal{L}$, a *chain* is the juxtaposition of symbols of $\mathcal{L}$. More mathematically, a chain is a sequence (finite or infinite) of elements of $\mathcal{L}$. To state that two chains α_1, α_2 are equivalent we use $\equiv$ as $\alpha_1 \equiv \alpha_2$.

Examples of chains are $x_1x_2x_3\neg$, $\neg \wedge x_1(((($ and $(x_{43}R_{111}^{32} \wedge (f_{990}^1))\rightarrow f_1^{90}R_1^2R_1^{3132}f_2^0\wedge$. As you may see, these chains are nonsense. Well, to construct any chain we have the liberty of putting any symbol of $\mathcal{L}$ after another. So we need rules, so these silly chains are excluded and only the ones which we want to develop more theory with remain.

Definition. Next are given the definitions of *term* and *formula*.

1. Any variable is a term.
2. Any constant symbol is a term.
3. If $f_i^{m_i}$ is a function symbol and $t_1, \dots, t_{m_i}$ are terms, then $f_i^{m_i}(t_1, \dots, t_{m_i})$ is a term.
4. For a relation symbol $R_j^{n_j}$ and terms $t_1, \dots, t_{n_j}$ the chain $R_j^{n_j}(t_1, \dots, t_{n_j})$ is a formula. Moreover, it is an *atomic formula*.
5. For a formula φ , $\neg\varphi$ is a formula.
6. For formulas φ and ψ , $\varphi \rightarrow \psi$ is a formula.
7. For formulas φ and ψ , $\varphi \wedge \psi$ is a formula.
8. For formulas φ and ψ , $\varphi \vee \psi$ is a formula.
9. For formulas φ and ψ , $\varphi \leftrightarrow \psi$ is a formula.
10. For a formula φ and a variable x_l , $\forall x_l(\varphi)$ is a formula.
11. For a formula φ and a variable x_l , $\exists x_l(\varphi)$ is a formula.
12. These are the only rules for the construction of terms and formulas.

We won't prove unique readability, because we are not defining things in a formal enough way yet, and because there's no unique readability for this way of constructing formulas. So, for formulas α, β and γ we can form ambiguous formulas like $\alpha \rightarrow \beta \rightarrow \gamma$ where we have two possible readings. To avoid this misunderstanding, we place parenthesis where its appropriate.

We interpret as usually. $\forall x$ as “for all x ”, $\exists x$ as “an x exists” and so on.

Here are some examples for what this definition can and cannot express. I will omit indexes indicating range, assuming that the reader understands that this doesn't matter because its a justified abuse of notation.

Example 1.1: Let $\mathcal{L}_{f_x}$ be the formal language with one relator symbol of range 2, $=$, no constant symbols and an infinite amount of function symbols. Call this language the language of evaluations. Then, the following are formulas of this language.

1. $\forall x(=(g(f(x), x), h(x, y)))$
2. $=(f(x), f(y)) \rightarrow =(x, y)$
3. $\exists x(\forall y(=(x, y)))$
4. $=(f(x), k(x)) \wedge =(k(x), l(x)) \rightarrow =(f(x), l(x))$

One can verify if a chain is a formula using the definition as follows:

- s1. We have the chain $\forall x(=(g(f(x), x), h(x, y)))$.
- s2. For it to be a formula, $=(g(f(x), x), h(x, y))$ should be a formula.
- s3. Then both $g(f(x), x)$ and $h(x, y)$ should be terms.
- s4. So $f(x), x, y$ should be terms.
- s5. Finally, x, y are variables, therefore terms, and so the whole chain is a formula.

You can also draw a tree for a chain using the formation rules.

Some general tree

To not lose generality the symbol $=$ isn't treated any special. However, we know this guy is representing the equality. So for an exception for it and any other known relation or function symbols, we write any privileged range 2 symbol $\mathcal{S}$ in a formula as $x\mathcal{S}y$ for terms x and y .

Example 1.2: Let $\mathcal{L}_A$ be the formal language with constants 0, function symbols of range 2: $+$, $\cdot$ and S and the relation symbol $=$. Call this language the formal language of arithmetic. Then, the following formulas are named the Robinson's arithmetic axioms.

1. $\forall x(\neg S(x) = 0)$
2. $\forall x(\forall y(S(x) = S(y) \rightarrow x = y))$
3. $\forall y(y = 0 \vee \exists x(S(x) = y))$
4. $\forall x(x + 0 = x)$
5. $\forall x(\forall y(x + S(y) = S(x + y)))$
6. $\forall x(x \cdot 0 = 0)$
7. $\forall x(x \cdot S(y) = x \cdot y + y)$

Remember the “first order” part of the definition? If not, go back. If yes, then assume that we now, by some divine power, are allowed to introduce chains of the next kind: $\forall\varphi(\Psi)$. Here, φ is any formula and Ψ is “second order” formula (the aforementioned chain is also one), that is, it may contain formulas as variables⁴. Now, if we interpret $\varphi(x)$ as in stating a property (that I soon will define) then the following is the (second order) known principle of induction:

$$8_2. \forall\varphi((\varphi(0) \wedge (\varphi(x) \rightarrow \varphi(S(x)))) \rightarrow \forall x(\varphi(x)))$$

If we drop the fancy act we get a schema for any property φ , still in the first order language.

$$8_1. (\varphi(0) \wedge (\varphi(x) \rightarrow \varphi(S(x)))) \rightarrow \forall x(\varphi(x))$$

that then makes of this collection of formulas an infinite one (one formula for each property). So our first order formal language definition can't describe induction without running in either of these problems.

Example 1.3: Let $\mathcal{L}_{\text{日本語}}$ be the formal language consisting of constant symbols denoted as japanese nouns 猫 (cat), 魚 (fish), 数学 (math), 数学者 (mathematician), ..., variables interpreted as “events” $e, e_n, \dots$ ⁵, unary relation symbols denoted by verbs 食べる (to eat), 見

⁴Therefore, it quantifies over formulas of the first order language. We can take this further and permit new chains to quantify over second order formulas, creating a third order language, and so on. However, bringing new conflicts.

⁵This is called neo-Davidsonian event semantics. See [MP11] if you are interested.

る (to see), 勉強する (to study), なる (to become), である (to be⁶), ... and binary relation symbols as grammatical particles が (agent), を (patient), に (target), で (context), ... No function symbols. Call this, a deficient attempt to formalize the japanese language.

Then, the following are formulas of $\mathcal{L}_{\text{日本語}}$:

1. $\exists e(\text{食べる}(e) \wedge \text{が}(\text{猫}, e) \wedge \text{を}(\text{魚}, e))$
2. $\exists e(\text{見る}(e) \wedge \text{が}(\text{魚}, e) \wedge \text{を}(\text{猫}, e))$
3. $\exists e(\text{勉強する}(e) \wedge \text{が}(\text{猫}, e) \wedge \text{を}(\text{数学}, e))$

The first one is interpreted as: “There is an event, to eat is related to the event, the agent of the event is a cat and the patient of the event is a fish”. In dictionary japanese, a written form could be the next.

TL1. 猫が魚を食べる. The cat eats the fish.

TL2. 魚が猫を見る. The fish sees the cat.

TL3. 猫が数学を勉強する. The cat studies mathematics.

Now, if we are considering on trying to make a universal assertion, then we would use the quantifier $\forall$ to express it. Say that we want to express “every cat is not a mathematician” in $\mathcal{L}_{\text{日本語}}$. We decompose it as “For any noun that is cat, cat is not mathematician”, but we find a problem, the quantifier requires a variable, and in this formal language, variables are events. Therefore we would state something for any event, then, “for any event that is cat ...”. Absurd. To solve this in a lazy way, assume that variables can be separated as nouns and events, i.e., there are formulas that express that a variable is a noun and a variable is an event, $\text{No}(x)$ and $\text{Ev}(x)$. Then, we can construct the formula

- a. $\forall x(\forall e_1((\text{No}(x) \wedge \text{Ev}(e_1) \wedge \text{である}(e_1) \wedge \text{が}(x, e_1) \wedge \text{で}(\text{猫}, e_1)) \rightarrow \neg \exists e_2(\text{Ev}(e_2) \wedge \text{である}(e_2) \wedge \text{が}(x, e_2) \wedge \text{で}(\text{数学者}, e_2))))$

that translates to

TLa. どの猫も数学者ではない. No cat is a mathematician.

Next, consider the formula

- i. $\exists e(\text{Ev}(e) \wedge \text{No}(x) \wedge \text{なる}(e) \wedge \text{が}(x, e) \wedge \text{を}(\text{魚}, e))$

which translates to any of the following

TLi. 魚になる. I become fish.

TLii. 魚になる. Cat becomes fish.

TLiii. 魚になる. Become fish.

As you can notice, not only absurd sentences can be formulas but also the complexity of our formal language greatly exceeds that of the natural one. This is because of the reliance on context that japanese language commonly uses. Not only this language, but also english, and some others. The usually prescind of important information that needs to be filled up in the formal expression.

Example 1.4: For categories, we consider arrows or functions as having a domain, codomain and its name. Here we'll try to design a formal language for variables that consists of three

⁶This is used to state a fact. But we use it as the “to be” verb.

things. First, note that the identity, and in general any permutation, can represent a collection X . So, for any function $f : A \rightarrow B$ we can establish $A = \text{id}_A$ and $B = \text{id}_B$. So here, to only work with the same kind of variable, we write $f : \text{id}_A \rightarrow \text{id}_B$. Abstractly, for arrows, an object will be determined by an arrow a if we have $\text{dom}(a) = a$ and $\text{cod}(a) = a$. Let $\mathcal{L}_C$ be the formal language consisting of unary function symbols dom and cod , a range $\exists$ relation symbol $\circ$ and the $=$ symbol. We shall call this the language of functions. The variables are denoted, for now, with $f, g, h, \dots$. We put $f = g \circ h \equiv \circ(f, g, h)$.

Then we have the following definition of category found in [nLa26]:

1. $\forall f(\text{dom}(\text{dom}(f)) = \text{dom}(f) \wedge \text{dom}(f) = \text{cod}(\text{dom}(f)))$
2. $\forall f(\text{cod}(\text{cod}(f)) = \text{cod}(f) \wedge \text{dom}(f) = \text{dom}(\text{cod}(f)))$
3. $\forall f(\forall g(\text{dom}(g) = \text{cod}(f) \leftrightarrow \exists h(h = g \circ f)))$
4. $\forall(h \forall f(\forall g(h = g \circ f \rightarrow (\text{dom}(h) = \text{dom}(f) \wedge \text{cod}(h) = \text{cod}(g))))))$
5. $\forall f(f = f \circ \text{dom}(f)) \wedge f = \text{cod}(f) \circ f$
6. $\forall f(\forall g(\forall h(\forall k(\forall l(\forall \alpha(\forall \beta((k = g \circ f \wedge \alpha = h \circ k \wedge l = h \circ g \wedge \beta = l \circ f) \rightarrow \alpha = \beta)))))))$

Interpreting these axioms, we can see that $\text{dom}(f)$ and $\text{cod}(f)$ are what we call objects. We shall then define: $\text{obj}(f) \equiv \text{dom}(f) = f \wedge \text{cod}(f) = f$.

The following are some “categorical axioms for sets” due to [Law64].

- a. $\exists t(\text{obj}(t) \wedge \forall a(\text{obj}(a) \rightarrow \exists f(\text{dom}(f) = a \wedge \text{cod}(f) = t \wedge \forall g((\text{dom}(g) = a \wedge \text{cod}(g) = t) \rightarrow f = g))))$
- b. $\exists i(\text{obj}(i) \wedge \forall b(\text{obj}(b) \rightarrow \exists f(\text{dom}(f) = i \wedge \text{cod}(f) = b \wedge \forall g((\text{dom}(g) = i \wedge \text{cod}(g) = b) \rightarrow f = g))))$
- c. $\forall a(\forall b((\text{obj}(a) \wedge \text{obj}(b)) \rightarrow \exists c(\exists \pi_a(\exists \pi_b(\text{obj}(c) \wedge ((\text{dom}(\pi_a) = c \wedge \text{cod}(\pi_a) = a) \wedge (\text{dom}(\pi_b) = c \wedge \text{cod}(\pi_b) = b) \wedge \forall x(\forall f(\forall g((\text{dom}(f) = x \wedge \text{cod}(f) = a \wedge \text{dom}(g) = x \wedge \text{cod}(g) = b) \rightarrow \exists h(\text{dom}(h) = x \wedge \text{cod}(h) = c \wedge \circ(f, \pi_a, h) \wedge \circ(g, \pi_b, h) \wedge \forall k((\text{dom}(k) = x \wedge \text{cod}(k) = c \wedge \circ(f, \pi_a, k) \wedge \circ(g, \pi_b, k)) \rightarrow h = k))))))))))$

Our formation rules and conventions are not enough to avoid constructing such monstrous chains. To simplify this a bit more, we can identify a pattern in all formulas. This pattern should be known to the reader and is usually abbreviated by the chain $\exists!$. For any “property” that presents in a formula as $\exists x(\varphi(x) \wedge \forall y(\varphi(y) \rightarrow x = y))$ we abbreviate by $\exists!x(\varphi(x))$. Then a more compact and elegant form is obtained by using this shorthand and defining the next chains:

$$\text{prodd}(a, b, \pi_a, \pi_b, c) \equiv \text{obj}(c) \wedge \text{dom}(\pi_a) = c \wedge \text{dom}(\pi_b) = c \wedge \text{cod}(\pi_a) = a \wedge \text{cod}(\pi_b) = b$$

$$\text{prodp}(a, b, \pi_a, \pi_b, c) \equiv \forall x(\forall f(\forall g((\text{dom}(f) = x \wedge \text{dom}(g) = x \wedge \text{cod}(f) = a \wedge \text{cod}(g) = b) \rightarrow \exists!h(\text{dom}(h) = x \wedge \text{cod}(h) = c \wedge f = \pi_a \circ h \wedge g = \pi_b \circ h))))$$

- a. $\exists t(\text{obj}(t) \wedge \forall a(\text{obj}(a) \wedge \exists!f(\text{dom}(f) = a \wedge \text{cod}(f) = t)))$
- b. $\exists i(\text{obj}(i) \wedge \forall b(\text{obj}(b) \wedge \exists!f(\text{dom}(f) = i \wedge \text{cod}(f) = b)))$
- c. $\forall a(\forall b((\text{obj}(a) \wedge \text{obj}(b)) \rightarrow \exists c(\exists \pi_a(\exists \pi_b(\text{prodd}(a, b, \pi_a, \pi_b, c) \wedge \text{prodp}(a, b, \pi_a, \pi_b, c))))))$

If you haven't figured out what they are meant to interpret, 1. and 2. refer to terminal and initial objects, that is, objects such that for any morphism or arrow f with codomain t (domain i) such arrow is unique. The third one refers to the existence of products along with its universal property.

Example 1.5: Let $\mathcal{L}_{LM}$ be the language consisting of an infinite amount of unary function symbols indexed by a ring as $f_r, f_s, \dots$, along with a binary symbol $+$, the relation $=$ and a constant symbol 0 . Call this the formal language of left-modules. Denote a term formed with a unary function symbol $f_r(x)$ by $f_r x$. Then, the axioms for left-modules are as shown below:

1. $\forall x(\forall y(\forall z((x + y) + z = x + (y + z))))$
2. $\forall x(x + 0 = x)$
3. $\forall x(\exists y(x + y = 0))$
4. $\forall x(\forall y(x + y = y + x))$
- 5_s. $\forall x(\forall y(f_r(x + y) = f_r x + f_r y))$
- 6_s. $\forall x(f_{r+s} x = f_r x + f_s x)$
- 7_s. $\forall x(f_r f_s x = f_{rs} x)$

Notice that formulas from 5 to 7 are schemes for any unary function symbols f_r, f_s , similar to the Robinson's arithmetic example.

Another interesting examples are left as an exercise to the reader. One question that may come to anyone is “why complicate things so much?” The answer is just because, *tehe~*. A second one is “why don't you use multiple variables to represent different things?” That is because we don't know a precise way to interpret the quantifiers as taking elements of multiple collections to assert something. Like in the Example 1.4 we can't use variables to represent both objects and arrows, because then, according to our rules, the formulas stated also hold for objects, so then objects satisfy associativity and there's a composition of objects. This here, requires more machinery to distinguish between multiple collections of variables. We call this abstract place, from where we take objects to quantify, the *universe of discourse*. A formal language with only one universe of discourse is called a *single-sorted*, *one-sorted* or *unsorted* formal language. A formal language with multiple universes of discourse is called a *many-sorted* formal language. To formalize categories we can use something called *first order logic of dependent sorts* which requires that we talk about types. This is interesting and will be treated shortly further in this text, but, for now, let's assume one universe of discourse.

The time has come. Define the following.

Definition. A *free variable* of a formula φ is a variable x not quantified. That is, not in the scope of a quantifier $\forall x$ or $\exists x$. The next definition via recursion is given:

1. If φ is an atomic formula, then the free variables of φ are all its variables.
2. If $\varphi \equiv \neg\psi$ then the free variables of φ are the free variables of ψ .
3. If $\varphi \equiv \psi_1 \wedge \psi_2$ then the free variables of φ are the free variables of both ψ_1 and ψ_2 .
4. If $\varphi \equiv \psi_1 \vee \psi_2$ then the free variables of φ are the free variables of both ψ_1 and ψ_2 .
5. If $\varphi \equiv \psi_1 \rightarrow \psi_2$ then the free variables of φ are the free variables of both ψ_1 and ψ_2 .
6. If $\varphi \equiv \psi_1 \leftrightarrow \psi_2$ then the free variables of φ are the free variables of both ψ_1 and ψ_2 .
7. If $\varphi \equiv \forall x\psi$ then the free variables of φ are the free variables of ψ without x .
8. If $\varphi \equiv \exists x\psi$ then the free variables of φ are the free variables of ψ without x .

Definition. A *bound variable* of a formula φ is a quantified variable. Recursively:

1. If φ is an atomic formula, it has no bound variables.
2. If $\varphi \equiv \neg\psi$ then its bound variables are the bound variables of ψ .
3. If $\varphi \equiv \psi_1 \wedge \psi_2$ then its bound variables are both the bound variables of ψ_1 and ψ_2 .
4. If $\varphi \equiv \psi_1 \vee \psi_2$ then its bound variables are both the bound variables of ψ_1 and ψ_2 .
5. If $\varphi \equiv \psi_1 \rightarrow \psi_2$ then its bound variables are both the bound variables of ψ_1 and ψ_2 .
6. If $\varphi \equiv \psi_1 \leftrightarrow \psi_2$ then its bound variables are both the bound variables of ψ_1 and ψ_2 .
7. If $\varphi \equiv \forall x\psi$ then its bound variables are the bound variables of ψ along with x .
8. If $\varphi \equiv \exists x\psi$ then its bound variables are the bound variables of ψ along with x .

And to be a bit more precise.

Definition. A *subformula* ψ of a formula φ is defined inductively as:

1. If φ is an atomic formula then φ is its only subformula.
2. If $\varphi \equiv \neg\psi$ then its subformulas are $\neg\psi$ and the subformulas of ψ .
3. If $\varphi \equiv \psi_1 \wedge \psi_2$ then its subformulas are φ and the subformulas of both ψ_1 and ψ_2 .
4. If $\varphi \equiv \psi_1 \vee \psi_2$ then its subformulas are φ and the subformulas of both ψ_1 and ψ_2 .
5. If $\varphi \equiv \psi_1 \rightarrow \psi_2$ then its subformulas are φ and the subformulas of both ψ_1 and ψ_2 .
6. If $\varphi \equiv \psi_1 \leftrightarrow \psi_2$ then its subformulas are φ and the subformulas of both ψ_1 and ψ_2 .
7. If $\varphi \equiv \forall x\psi$ then its subformulas are φ and the subformulas of ψ .
8. If $\varphi \equiv \exists x\psi$ then its subformulas are φ and the subformulas of ψ .

Example 1.6: Taking formulas of the examples showed before, we identify free and bound variables.

E1, 1. $\forall x(g(f(x), x) = h(x, y))$ has x as a bound variable and y as a free one.

E1, 4. $(f(x) = k(x) \wedge k(x) = l(x)) \rightarrow f(x) = l(x)$ has no bound variables.

E2, 1. $\forall x(\neg S(x) = 0)$ has the bound variable x .

E3, 4. The subformula $x + 0 = x$ has no bound variables. Whilst the formula has x .

E3, TLiii. 魚になる has the free variables なる and x .

E3, 8₂. $(\varphi(0) \wedge (\varphi(x) \rightarrow \varphi(S(x)))) \rightarrow \forall x(\varphi(x))$. Here x is both a free and bound variable.

E4, 1. $\forall f(\text{dom}(\text{dom}(f)) = \text{dom}(f) = \text{cod}(\text{dom}(f)))$ has the bound variable f .

E4, 3. The subformula $\exists h(h = g \circ f)$ has g and f as free variables and h as a bound one. The formula has no free variables.

E5, 2. $\forall x(x + 0 = x)$ has the bound variable x .

You might have noticed by now, but here are the definitions of property and statement.

Definition. A *property* of x is a formula with a free variable x . If φ has a free variable x then we write $\varphi(x)$. If multiple variables $x_1, \dots, x_n$ occur free in a formula ψ then we write $\psi(x_1, \dots, x_n)$ and say that ψ is a property of $x_1, \dots, x_n$.

A *statement* is a formula with no free variables.

Therefore, every formula given on $\mathcal{L}_{LM}, \mathcal{L}_A$ and $\mathcal{L}_C$ were statements. This works just fine for what they were meant to represent, axioms. Now, that doesn't quite work for the formula $x + 0 = 0$, which is just saying something about 0 and some x , but not about any variable x .

This concludes the syntactics of formal languages.

Now, to the semantics. We designed formal languages according to certain things we had in mind. Not only that, certain structures. Statements like $\forall x(\exists y(x + y = 0))$ are accurately describing rings, modules, vector spaces, but they do not describe a property of natural numbers. This, will be our notion of validness. A statement is valid for a structure if it correctly expresses its properties. To formalize this, let us define the following.

Definition. For a formal language $\mathcal{L}$, a $\mathcal{L}$ -*structure* $\mathfrak{A}$ consists of a collection U called a universe along with an interpretation of each type of symbol in $\mathcal{L}$. This is:

1. The interpretation assigns an m -ary function in U for each function symbol f^m .
2. It assigns an n -ary relation in U to every relation symbol R^n .
3. It assigns an element of U to any constant symbol.

When we think of this definition and the universe as classes or sets, we then get a mathematical formalization of what a structure is.

The examples shown before are modeled after an structure, so there's a structure for each of them. Let's see this more precisely.

Example 1.7: 1. For $\mathcal{L}_A$, we choose the natural numbers as the universe. The structure $\mathfrak{N}$ is then described as following. We assign $S(n)$ to the function that outputs the successor of a natural number n , described by $n + 1$. For $+$ and $\cdot$ we do the obvious. Same goes for constant 0. Then, Robinson's axioms, that is, the statements constructed are intuitively true for the structure $\mathfrak{N}$ we have given.

To make this more simple, let us denote the structure $\mathfrak{N}$ as $\mathfrak{N} = \langle \mathbb{N}, n \mapsto n + 1, +, \cdot, 0 \rangle$ and the following structures.

2. For $\mathcal{L}_{\text{日本語}}$ we choose the japanese language as $\mathfrak{J} = \langle \text{nouns, verbs, が, に, で, を} \rangle$. Here, the structure is just as modeled. If we try with the english language, then we would use $\mathfrak{E} = \langle \text{nouns, verbs, is, the one..., towards, by means of, does} \rangle$. The examples given on this formal language are then translated to its structure. Statements are determined true or false intuitively depending on what it is referring to in the natural language.
3. For $\mathcal{L}_C$, if we use naive set theory, we have $\mathfrak{S} = \langle X^Y, D : X^Y \rightarrow X^Y, C : X^Y \rightarrow X^Y, \{(f, g, h) \mid C(f) = D(g) \wedge h = f \circ g\} \rangle$, where $D(f)$ is the identity on the domain of f and $C(f)$ is the identity in the codomain of f . This structure intuitively satisfies the axioms given. Then it formally forms a one-sorted category.
4. For $\mathcal{L}_{LM}$ any known left-module suffices. Take, for example, real numbers and the ring $\mathbb{Z}$.

Exercises

1. Design and describe a first order formal language for a complete ordered field. Then, state the axioms.
2. Design and describe a first order formal language for vector spaces. Then state the axioms.

1.2 Propositional logic

1.3 First order logic

1.4 Non-classical logic

2

Introduction to set theory

2.1 Elements of set theory

2.2 Relations and functions

2.3 Natural numbers

2.4 Integers and rationals

2.5 Real numbers

2.6 Ordinals

2.7 Cardinality

2.8 Choice axiom

2.9 Well foundedness

3

More set theory

3.1 Classes and sets

3.2 Limits of infinity

3.3 Universes

3.4 Constructibility

3.5 Further topics on set theory

4

Structures

4.1 Historical note

4.2 Sets as the foundation

4.3 Universal algebra

4.4 Introduction to categories

5

Foundations of mathematics

5.1 Category theory

5.2 Metamathematics

5.3 Mathematical logic

5.4 Type theory

5.5 Philosophical concerns

Bibliography

- [Law64] F. William Lawvere. *An elementary theory of the category of sets*. University of Chicago, 1964.
- [MP11] von Heusinger Maienborn **and** Portner. “Event semantics”. **in**(2011).
- [nLa26] nLab authors. *single-sorted definition of categories*. <https://ncatlab.org/nlab/show/single-sorted+definition+of+categories>. Revision 10. **september** 2026.